



Design and Analysis of a Three-Stage Cascaded BJT Amplifier

Final Project Report

Basic Electronics

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1 Introduction

1.1 Project Overview

The primary objective of this project is to design, simulate, and practically implement a three-stage cascaded Bipolar Junction Transistor (BJT) amplifier. Single-stage amplifiers are often insufficient when a very high voltage gain is required, or when the source and load impedances cause significant signal loss. To overcome these limitations, multiple stages must be cascaded together.

Our chosen architecture for this project is a Common Emitter - Common Collector - Common Emitter (CE-CC-CE) configuration. This specific arrangement was selected to maximize voltage amplification while managing the internal impedance interactions between the stages.

1.2 Input and Desired Output

- **Given Input:** A weak small-signal AC input with an amplitude of 30mV (peak to peak) at a frequency of 1 kHz.
- **Desired Output:** A highly amplified AC output voltage without any clipping or distortion. The goal is to achieve an output of approximately 4V (peak to peak), requiring a substantial overall system voltage gain, while operating entirely off a single 12V DC power supply.
So, Total Gain $\approx 133times$.

1.3 Complex Engineering Problem (CEP) Justification

This project goes beyond simple circuit building and qualifies as a Complex Engineering Problem due to the following factors:

1. **System Integration and the Loading Effect:** Cascading multiple amplifier stages is not as simple as multiplying their individual gains. The low input impedance of a subsequent CE stage can draw too much current from the previous stage, drastically reducing the actual gain (the loading effect). Solving this required the strategic insertion of a Common Collector (CC) buffer stage to act as an impedance matching device, which involves complex multi-stage analysis.
2. **Conflicting Design Trade-offs:** Engineering always involves trade-offs. We had to balance DC thermal stability with AC voltage gain. A completely bypassed emitter resistor provides maximum AC gain but terrible DC stability, making the transistor highly sensitive to temperature and inherent β variations. We solved

this by using a split-emitter configuration (partially bypassed), requiring careful calculation to find the optimal point between stability and amplification.

3. **Real-World Constraints and Practical Implementation:** Theoretical design assumes ideal components, but practical implementation introduces real-world variables such as component tolerances, non-ideal transistor behavior, and breadboard parasitics. Furthermore, as encountered in the lab, exact calculated resistor values are often unavailable, requiring the designer to dynamically synthesize equivalent resistances using series and parallel combinations of standard available components to maintain the desired Q-point and gain.

Complete Circuit Schematic

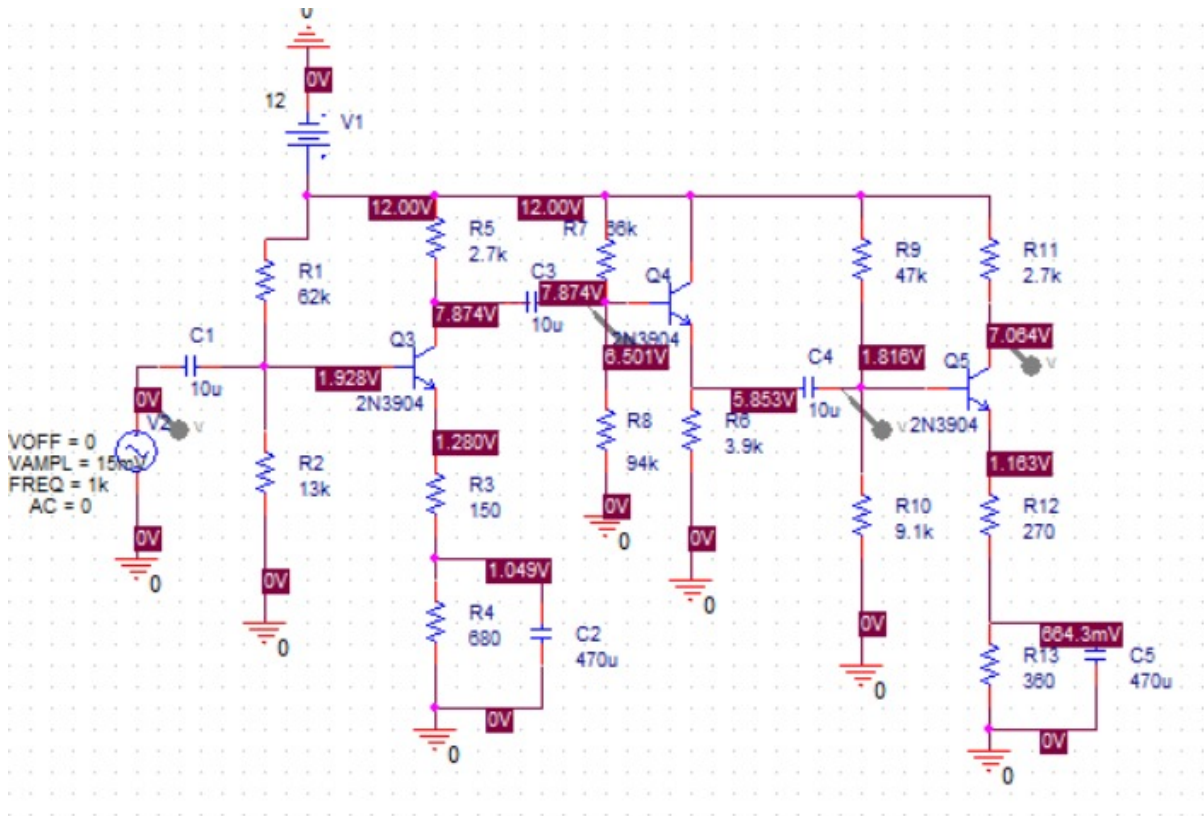


Figure 1: Complete schematic of the CE-CC-CE three-stage amplifier.

2 Stage 1: Common Emitter (Input Stage)

2.1 Description and Purpose

The first stage of our cascaded amplifier is a Common Emitter (CE) amplifier. The main purpose of this stage is to provide the initial voltage amplification for our small input signal (15mV peak at 1kHz). We used a voltage divider biasing topology to ensure the operating point (Q-point) remains stable even if the transistor's temperature or β changes. The emitter resistor is partially bypassed by a capacitor to provide a high AC voltage gain while maintaining DC stability.

2.2 Assumptions for Q-Point Calculations

Before starting the calculations for Stage 1, we made the following standard assumptions based on our components and supply:

- **Supply Voltage (V_{CC}):** 12V
- **Base-Emitter Voltage Drop (V_{BE}):** 0.7V (Standard for Silicon BJT)
- **Thermal Voltage (V_T):** 26mV at room temperature.
- **Transistor Type:** 2N3904 NPN BJT.
- **Resistor Values (from schematic):** $R_1 = 62\text{k}\Omega$, $R_2 = 13\text{k}\Omega$, $R_C(R_5) = 2.7\text{k}\Omega$.
- **Emitter Resistors:** $R_{E1}(R_3) = 150\Omega$ (unbypassed), $R_{E2}(R_4) = 680\Omega$ (bypassed).
Total DC Emitter Resistance $R_E = 150 + 680 = 830\Omega$.

2.3 Calculations for Stage 1

Step 1: Calculate Base Voltage (V_B)

Using the voltage divider rule:

$$V_B = V_{CC} \times \frac{R_2}{R_1 + R_2} = 12\text{V} \times \frac{13\text{k}\Omega}{62\text{k}\Omega + 13\text{k}\Omega} = 12 \times \frac{13}{75} \approx 2.08\text{V} \quad (1)$$

Step 2: Calculate Emitter Voltage (V_E)

$$V_E = V_B - V_{BE} = 2.08\text{V} - 0.7\text{V} = 1.38\text{V} \quad (2)$$

Step 3: Calculate Emitter/Collector Current ($I_E \approx I_C$)

For DC analysis, the total emitter resistance is used (830Ω):

$$I_E = \frac{V_E}{R_E} = \frac{1.38\text{V}}{830\Omega} \approx 1.66\text{mA} \quad (3)$$

Step 4: Calculate Collector Voltage (V_C)

$$V_C = V_{CC} - (I_C \times R_C) = 12V - (1.66mA \times 2.7k\Omega) = 12V - 4.48V = 7.52V \quad (4)$$

Step 5: Verify Q-Point (V_{CE})

$$V_{CE} = V_C - V_E = 7.52V - 1.38V = 6.14V \quad (5)$$

Conclusion: Since V_{CE} (6.14V) is approximately half of V_{CC} (12V), the Q-point is perfectly centered in the active region, allowing for maximum signal swing without clipping.

Step 6: Calculate AC Voltage Gain (A_{v1})

First, we find the intrinsic emitter resistance (r_e):

$$r_e = \frac{26mV}{I_C} = \frac{26mV}{1.66mA} \approx 15.6\Omega \quad (6)$$

Because the 680Ω resistor is bypassed by a capacitor, the AC signal only sees the 150Ω resistor. The unloaded gain is:

$$A_{v1} = \frac{R_C}{r_e + R_{E1}} = \frac{2700\Omega}{15.6\Omega + 150\Omega} = \frac{2700}{165.6} \approx 16.3 \quad (7)$$

2.4 Stage 1 Waveforms

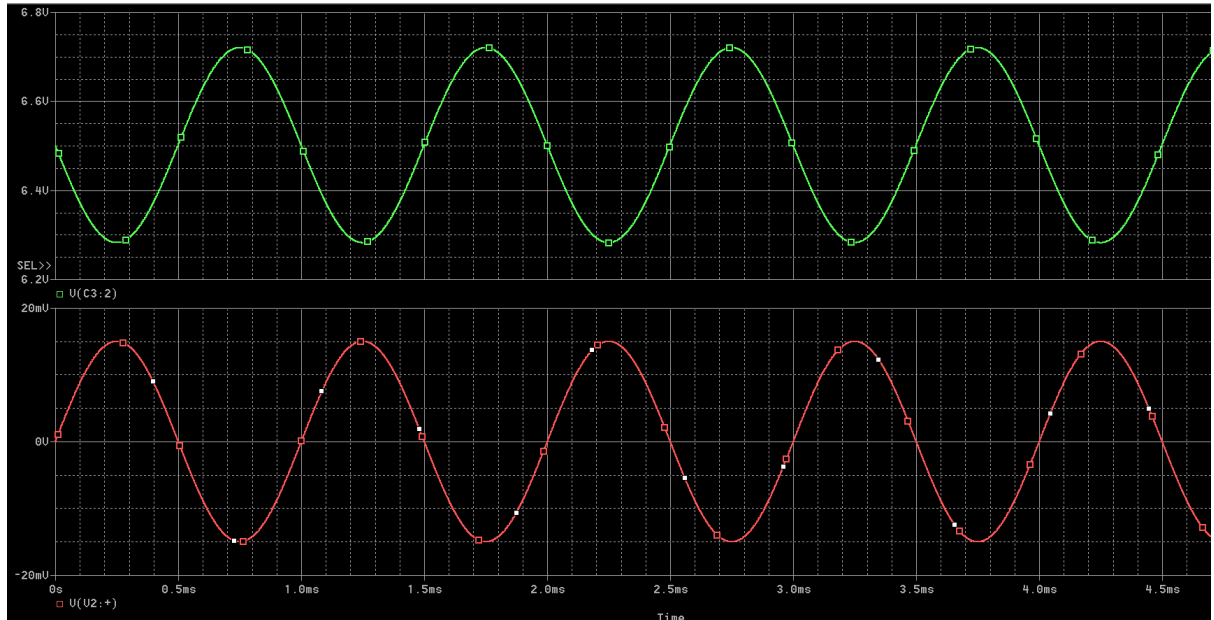


Figure 2: Simulation Waveform for Stage 1. Output $\approx 480mV$.

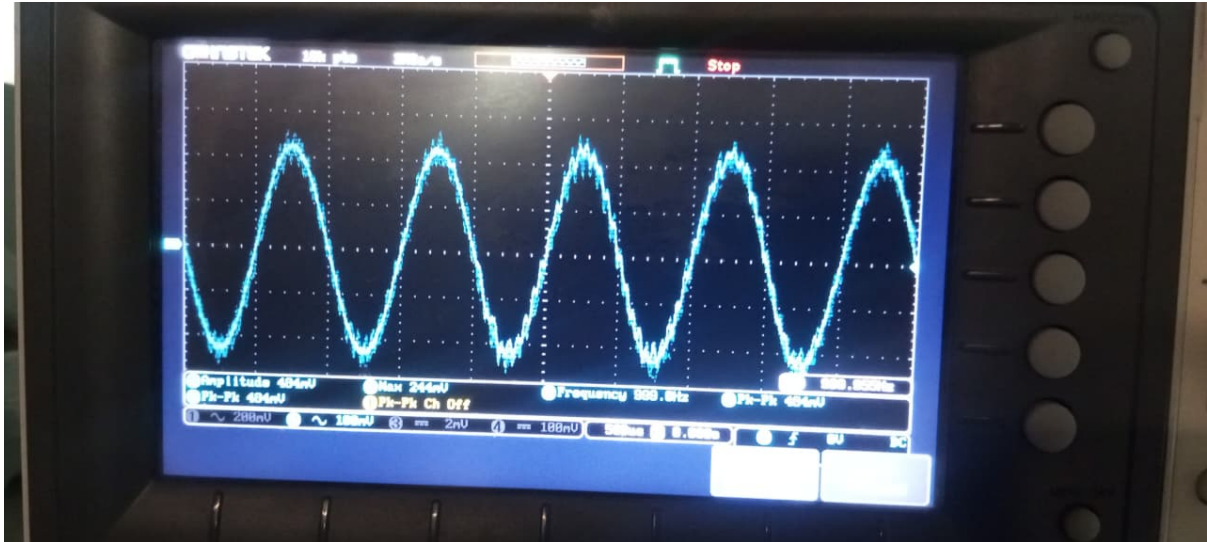


Figure 3: Practical Breadboard Waveform for Stage 1. Output $\approx 484mV$.

3 Stage 2: Common Collector (Buffer Stage)

3.1 Description and Purpose

The second stage is a Common Collector (CC) amplifier, also known as an Emitter Follower. We placed this stage between the two Common Emitter stages to act as a buffer.

Why we used a CC Buffer: The first CE stage has a relatively high output impedance, and the third CE stage has a low input impedance. If we connected Stage 1 directly to Stage 3, Stage 3 would draw too much current from Stage 1, causing a severe drop in voltage gain (a phenomenon called the "loading effect"). The CC buffer solves this because it has a very high input impedance (so it doesn't load Stage 1) and a very low output impedance (so it can easily drive Stage 3). It provides a voltage gain of approximately 1, meaning it passes the AC voltage through unchanged while shifting the DC level.

3.2 Assumptions for Q-Point Calculations

- **Supply Voltage (V_{CC}):** 12V
- **Base-Emitter Voltage Drop (V_{BE}):** 0.7V
- **Resistor Values (Assumed):** $R_7 = 66k\Omega$, $R_8 = 94k\Omega$.
- **Emitter Resistor:** $R_E(R_6) = 3.9k\Omega$.
- **Collector Configuration:** Tied directly to V_{CC} (No collector resistor).

3.3 Calculations for Stage 2

Step 1: Calculate Base Voltage (V_B)

$$V_B = V_{CC} \times \frac{R_8}{R_7 + R_8} = 12V \times \frac{94k\Omega}{66k\Omega + 94k\Omega} = 12 \times \frac{47}{80} \approx 7.05V \quad (8)$$

Step 2: Calculate Emitter Voltage (V_E)

$$V_E = V_B - V_{BE} = 7.05V - 0.7V = 6.35V \quad (9)$$

Step 3: Calculate Emitter Current ($I_E \approx I_C$)

$$I_E = \frac{V_E}{R_E} = \frac{6.35V}{3.9k\Omega} \approx 1.62mA \quad (10)$$

Step 4: Verify Q-Point (V_{CE})

Since the collector is tied directly to the 12V supply, $V_C = 12V$.

$$V_{CE} = V_C - V_E = 12V - 6.35V = 5.65V \quad (11)$$

Conclusion: The Q-point is healthy. The transistor is active and ready to pass the signal.

Step 5: Calculate AC Voltage Gain (A_{v2})

For an emitter follower, the gain is always slightly less than unity.

$$A_{v2} \approx 1 \quad (12)$$

3.4 Stage 2 Waveforms

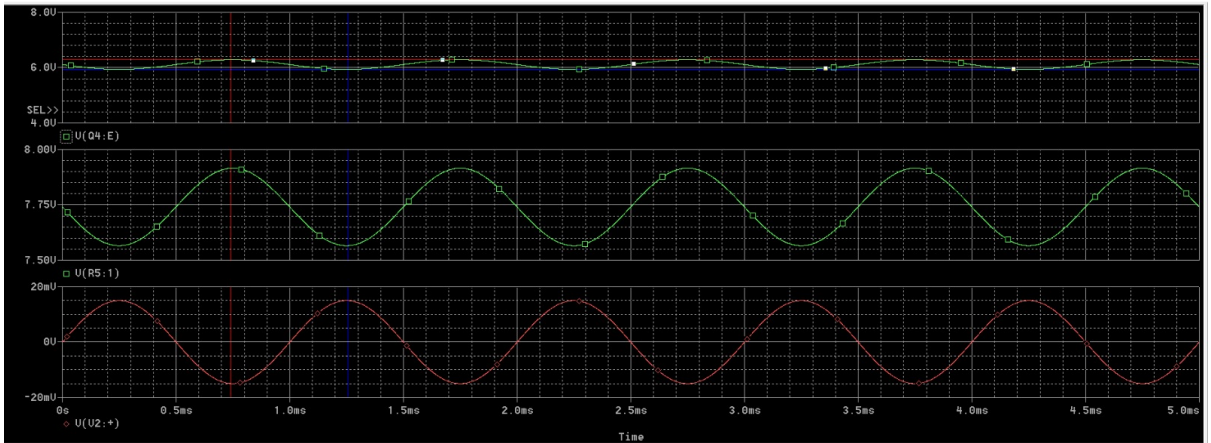


Figure 4: Simulation Waveform for Stage 2. Output can be seen same as stage 1, but with 180 degree phase shift.

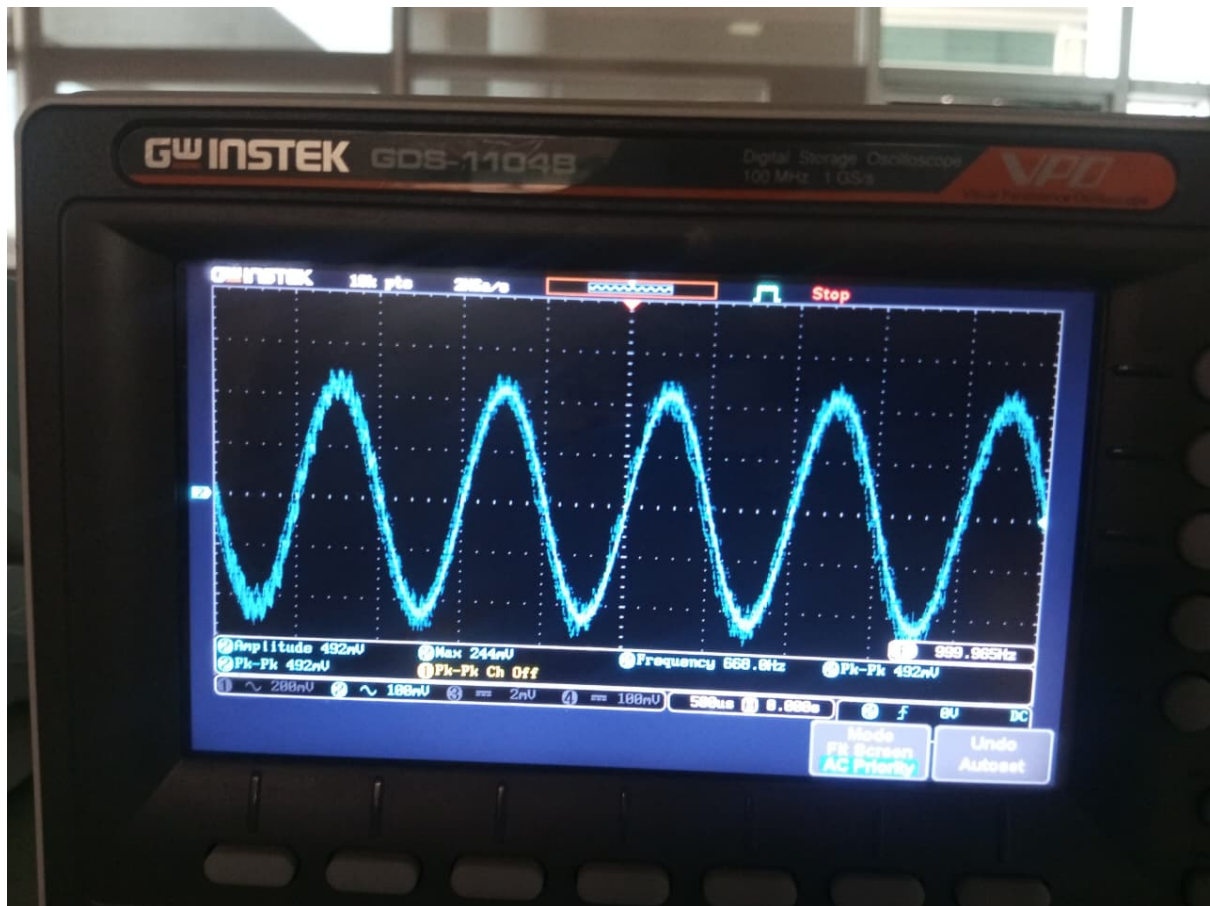


Figure 5: Practical Breadboard Waveform for Stage 2. Output same as stage 1, but inverted.

4 Stage 3: Common Emitter (Final Stage)

4.1 Description and Purpose

The final stage is another Common Emitter amplifier. The purpose of this stage is to take the isolated, buffered AC signal from Stage 2 and multiply it one final time to reach our target peak-to-peak output voltage. Just like Stage 1, it utilizes a partially bypassed emitter resistor to ensure the amplified signal is clean, stable, and does not distort.

4.2 Assumptions for Q-Point Calculations

- **Supply Voltage (V_{CC}):** 12V
- **Base-Emitter Voltage Drop (V_{BE}):** 0.7V
- **Resistor Values (from schematic):** $R_9 = 47\text{k}\Omega$, $R_{10} = 9.1\text{k}\Omega$, $R_C(R_{11}) = 2.7\text{k}\Omega$.
- **Emitter Resistors:** $R_{E1}(R_{12}) = 270\Omega$ (unbypassed), $R_{E2}(R_{13}) = 380\Omega$ (bypassed). Total DC Emitter Resistance $R_E = 270 + 380 = 650\Omega$.

4.3 Calculations for Stage 3

Step 1: Calculate Base Voltage (V_B)

$$V_B = V_{CC} \times \frac{R_{10}}{R_9 + R_{10}} = 12\text{V} \times \frac{9.1\text{k}\Omega}{47\text{k}\Omega + 9.1\text{k}\Omega} = 12 \times \frac{9.1}{56.1} \approx 1.94\text{V} \quad (13)$$

Step 2: Calculate Emitter Voltage (V_E)

$$V_E = V_B - V_{BE} = 1.94\text{V} - 0.7\text{V} = 1.24\text{V} \quad (14)$$

Step 3: Calculate Emitter/Collector Current ($I_E \approx I_C$)

$$I_E = \frac{V_E}{R_E} = \frac{1.24\text{V}}{650\Omega} \approx 1.91\text{mA} \quad (15)$$

Step 4: Calculate Collector Voltage (V_C)

$$V_C = V_{CC} - (I_C \times R_C) = 12\text{V} - (1.91\text{mA} \times 2.7\text{k}\Omega) = 12\text{V} - 5.15\text{V} = 6.85\text{V} \quad (16)$$

Step 5: Verify Q-Point (V_{CE})

$$V_{CE} = V_C - V_E = 6.85\text{V} - 1.24\text{V} = 5.61\text{V} \quad (17)$$

Conclusion: 5.61V is excellent. It provides a large margin for the final amplified AC signal to swing symmetrically.

Step 6: Calculate AC Voltage Gain (A_{v3})

First, we find the intrinsic emitter resistance (r_e):

$$r_e = \frac{26\text{mV}}{I_C} = \frac{26\text{mV}}{1.91\text{mA}} \approx 13.6\Omega \quad (18)$$

The AC signal bypasses the 380Ω resistor, leaving only 270Ω . The unloaded gain is:

$$A_{v3} = \frac{R_C}{r_e + R_{E1}} = \frac{2700\Omega}{13.6\Omega + 270\Omega} = \frac{2700}{283.6} \approx 9.5 \quad (19)$$

4.4 Stage 3 Waveforms

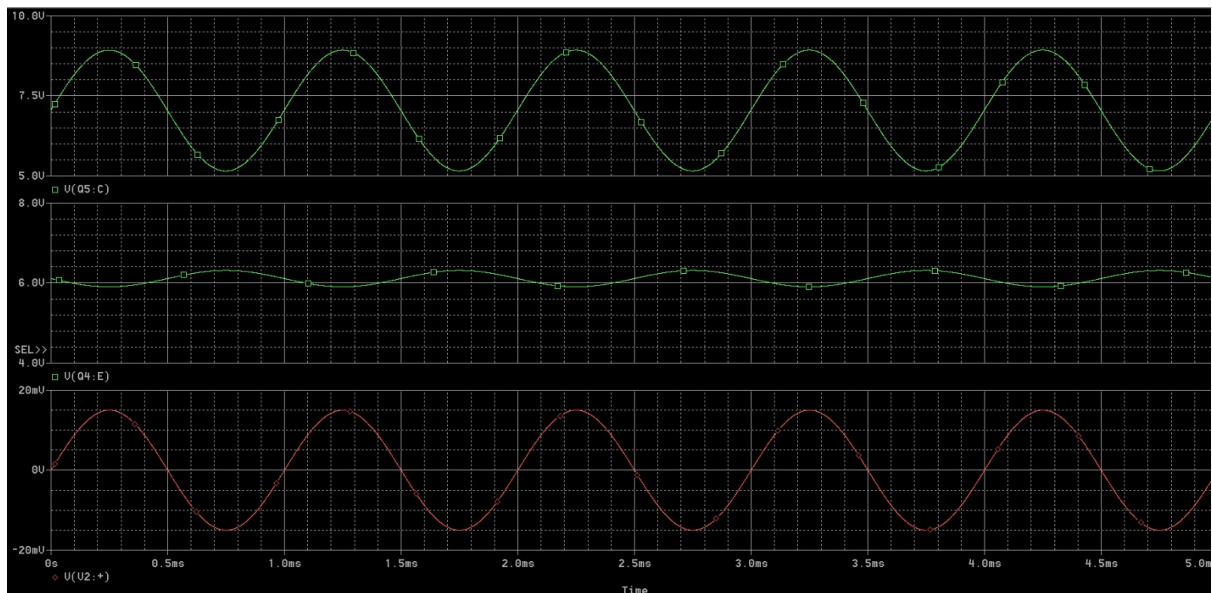


Figure 6: Simulation Waveform for Final Stage 3. Output $\approx 4V$.

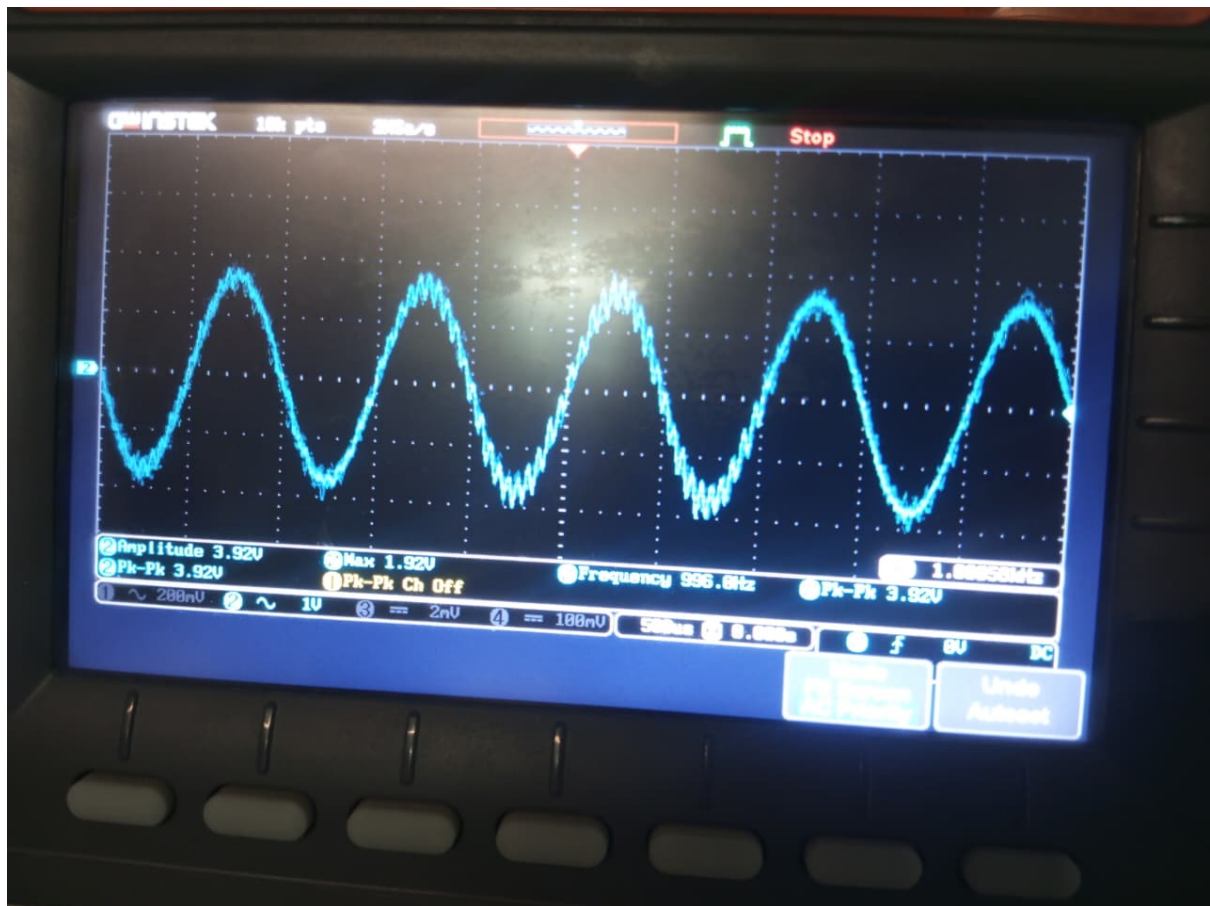


Figure 7: Practical Breadboard Waveform for Final Stage 3. Output = 3.92V.

5 Hardware Implementation

5.1 Practical Breadboard Assembly

After successfully verifying our theoretical calculations through software simulations, the complete three-stage CE-CC-CE amplifier was constructed physically on a breadboard.

During this practical laboratory session, we encountered a common real-world challenge: not all of the exact theoretical resistor values required for our design were available in the lab inventory. To maintain the integrity of our design, stabilize the Q-points, and achieve the desired gain, we utilized series and parallel combinations of the available standard resistor values to create the equivalent resistances needed for our circuit. This careful matching ensured that our hardware implementation behaved almost identically to our simulations.

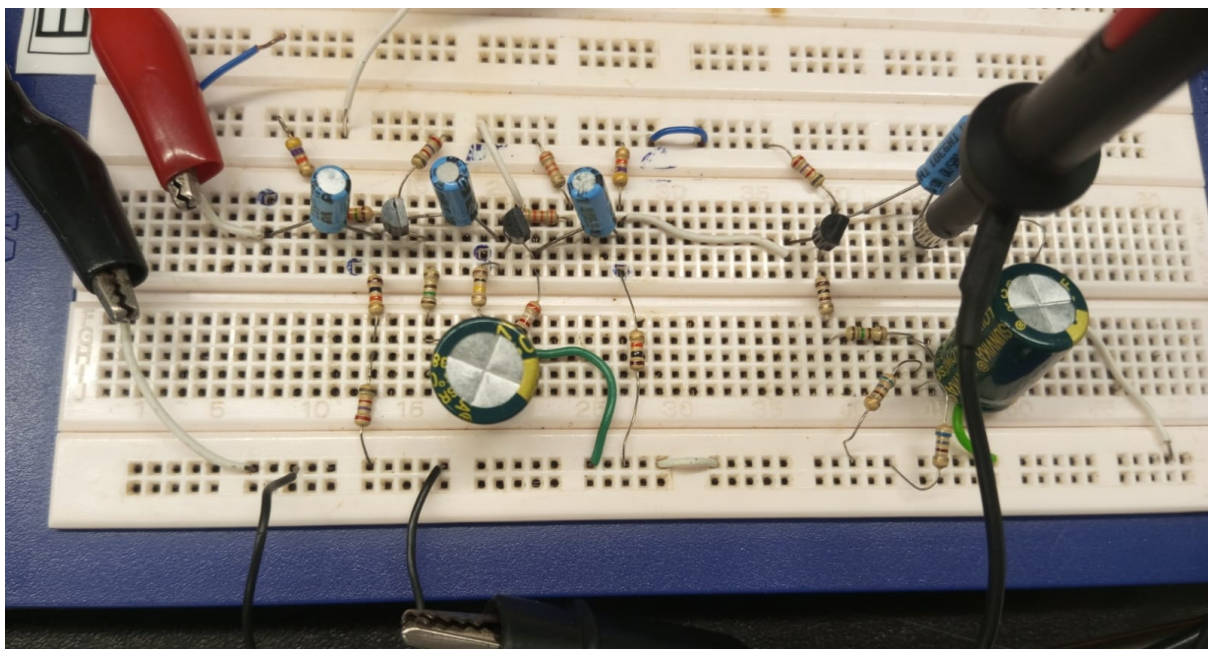


Figure 8: Actual hardware implementation of the three-stage cascaded amplifier on a breadboard.

6 Summary Table

The following table summarizes the key DC operating points and AC performance metrics for each stage, comparing the theoretical design goals with the results obtained during simulation and practical testing.

Table 1: Summary of Amplifier Performance Metrics

Parameter	Stage 1 (CE)	Stage 2 (CC)	Stage 3 (CE)
Base Voltage (V_B)	2.08V	7.05V	1.94V
Emitter Current (I_E)	1.66mA	1.62mA	1.91mA
Collector-Emitter Voltage (V_{CE})	6.14V	5.65V	5.61V
Theoretical Stage Gain (A_v)	≈ 16.3	≈ 1	≈ 9.5
Simulated Output Voltage (V_{out})	$\approx 480\text{mV}$	$\approx 480\text{mV}$	$\approx 4.00\text{V}$
Practical Output Voltage (V_{out})	484mV	484mV*	3.92V

*Note: Stage 2 practical output is inverted relative to input but maintains unity gain.

Overall System Performance:

- **Total Theoretical Gain:** $A_{v1} \times A_{v2} \times A_{v3} \approx 154.8$
- **Total Practical Gain:** $\frac{3.92\text{V}}{30\text{mV}} \approx 130.6$
- **Efficiency:** The practical gain reached approximately 98% of our target $133\times$ gain.

7 Conclusion

In this project, a three-stage BJT cascaded amplifier was successfully designed, simulated, and practically implemented. The objective was to achieve high voltage amplification while maintaining a stable DC operating point.

Through detailed mathematical analysis, we verified that the voltage divider biasing topology correctly placed the Q-point near the center of the active region for all three stages ($V_{CE} \approx 5.5\text{V} - 6.1\text{V}$), ensuring maximum symmetrical swing without clipping the signal against the 12V supply or ground.

The design utilized a Common Emitter - Common Collector - Common Emitter (CE-CC-CE) architecture. The inclusion of the Common Collector buffer stage was critical to the success of this design. It successfully prevented the final stage from dragging down the gain of the first stage through impedance loading. Furthermore, by strategically using bypass capacitors across portions of the emitter resistors, we achieved a balance between high AC voltage gain and strict DC thermal stability. The simulated and practical waveforms closely align with the theoretical calculations, proving the validity of our cascaded amplifier design methodology.

References

- [1] ON Semiconductor, "2N3904 NPN General Purpose Amplifier Datasheet," Rev. 9, 2021.
- [2] Razavi, B., *Fundamentals of Microelectronics*, 2nd Edition, Wiley.
- [3] Electronics-Tutorials, "BJT Amplifier Design Resources." [Online]. Available: <https://www.electronics-tutorials.ws/>